

Experimental Demonstration of AI in Performance Optimization of Short-Reach Optical Interconnects

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Abstract We demonstrate a unified platform that compares AI-assisted optimization strategies for short-reach IM/DD interconnects (transmitter pre-distortion, neural-network and look-up-table receivers, and end-to-end learning) through live simulations and remotely operated laboratory experiments with real-time performance visualization. ©2026 The Author(s)

Introduction

The continuous growth of data traffic driven by cloud services, artificial intelligence (AI), and distributed computing has significantly increased the performance requirements of short-reach optical interconnects (SROI), typically based on co-packaged optics (CPO) or near-packaged optics (NPO), within data centers. The same technology has the potential to play a key role in 6G radio-access networks, due to its ability to enable high capacity at low energy consumption [1]. In this context, intensity-modulation and direct-detection (IM/DD) systems remain a key enabling technology due to their cost-efficiency and scalability. However, recent trends in post-200G short-reach optics are pushing IM/DD systems toward higher per-lane rates, including 400-Gb/s-class lanes [2]. At these rates, performance is increasingly constrained by bandwidth limitations, noise, fiber dispersion, transceiver nonlinearities, and digital signal processing (DSP) complexity [3], motivating the exploration of advanced optimization techniques.

The adoption of these techniques is further motivated by the fact that configuring SROI requires the joint optimization of several parameters, e.g., the bias voltages of a segmented Mach-Zehnder modulator (MZM), or the gains of a multi-stage transimpedance amplifier (TIA), which is highly challenging. Recently, AI-based methods have emerged as a powerful tool to enhance optical system performance by enabling adaptive and data-driven optimization of transmitter and receiver functionalities [4]. Techniques such as machine learning-based equalization, nonlinear compensation, and end-to-end system optimization have demonstrated the potential to outperform traditional model-based approaches, particularly in scenarios where accurate analytical modeling be-

comes challenging. Despite these advances, a unified practical comparison of different AI-assisted optimization strategies under realistic implementation conditions is still missing. In particular, there is a need to evaluate how transmitter-side, receiver-side, and end-to-end learning techniques perform when deployed across simulation environments and experimental testbeds, considering practical constraints such as limited training data, hardware impairments, and system latency.

In this demo paper, we present a unified experimental and simulation-assisted platform for AI-assisted optimization of short-reach IM/DD optical interconnects. The demonstration brings together transmitter-side learning, receiver-side neural-network equalization, look-up-table (LUT) based detection, and end-to-end learning strategies within a common evaluation workflow. The platform shows how learning-based models are trained, implemented, and validated, and how different optimization strategies can be compared side by side through local simulations and remotely operated laboratory experiments with real-time visualization of performance metrics, providing a practical proof of concept for deploying AI-assisted optimization under realistic hardware and system constraints. The platform turns a comparison that is otherwise read off published curves into a hands-on, side-by-side view on live optical hardware.

AI-Assisted Optimization Techniques

All techniques are demonstrated on a common bipolar pulse-amplitude modulation (BPAM) IM/DD platform at 20 GBd, the configuration used in our recent comparative study [5]. Unlike conventional unipolar PAM, BPAM transmits both positive and negative amplitudes and, with twofold oversampling, recovers the sign information lost at the square-law photodiode, improving energy efficiency at modest additional receiver complexity [6].

The format was introduced and first demonstrated experimentally by some of the present authors [7], with only a small body of subsequent independent work [8]. While AI-assisted optimization of unipolar PAM-4 is by now well established [9, 10], its experimental implementation across the BPAM/DD chain (transmitter, receiver, and end-to-end) remains largely unexplored and is the subject of this demonstration.

Transmitter-side learning. Bench characterization at 20 GBd shows that the 6-bit digital-to-analog converter (DAC) is a dominant non-linear element: its combined quantization and memory effects contribute a substantial share of the experimental implementation penalty, motivating transmitter-side pre-distortion alongside the receiver and end-to-end methods that follow [11]. The pre-distorter is learned with an indirect, surrogate-based scheme [12]: a differentiable neural-network model of the DAC, fed with a sliding input window that spans the device memory, is first trained on measured input/output waveforms, and the pre-distorter is then optimized by back-propagating through this frozen surrogate so that the predicted output matches the undistorted target. Two pre-distorter realizations, a compact neural network and a memory polynomial [13], are *fairly trained* within the same gradient-based framework, the memory polynomial fitted by back-propagation through the surrogate rather than by its classical indirect-learning architecture, enabling a like-for-like comparison. Because the stage acts entirely at the transmitter, it is complementary to the methods below and appears in the demonstration as an optional pre-processing block whose effect on the transmitted eye can be toggled.

Receiver-side learning. At the receiver the twofold-oversampled BPAM waveform is processed by parallel odd/even pipelines. The baseline is a feed-forward adaptive equalizer (AE), two finite-impulse-response (FIR) filters whose taps are pre-optimized under a least-mean-squares criterion on training symbols, which is inexpensive but cannot compensate the nonlinear distortions of the electro-optic chain [6, 10]. A neural-network equalizer replaces the linear filter with a learned classifier on a context window of received samples, trained by supervised learning with cross-entropy on measured sequences so that linear and non-linear impairments are mitigated jointly; both compact (a few tens of parameters) and deeper (a few thousand parameters) realizations are considered, as demonstrated experimentally for BPAM in [14]. A look-up-table (LUT) detector associates two transmitted bits with a sequence of n received samples taking values in $\{1, 2, \dots, 2^{N_{\text{ADC}}}\}$, given an N_{ADC} -bit analog-to-digital converter (ADC). The LUT is trained offline, approaching a maximum-

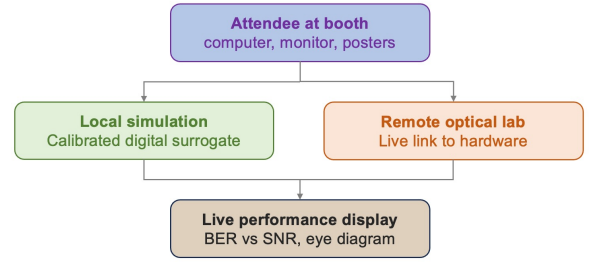


Fig. 1: Architecture of the demonstration.

a-posteriori probability (MAP) strategy, and the detection is optimal, for the same n and N_{ADC} , as long as the training is sufficiently accurate [5, 15]. The LUT has zero run-time computation and memory $2^{nN_{\text{ADC}}+1}$ bits. The AE, the neural-network equalizer, and the LUT together span a wide complexity/performance trade-off at the receiver.

End-to-end learning. The last group jointly optimizes transmitter, link, and receiver rather than treating them separately. The first model is an autoencoder-style transceiver in which each analog block of the IM/DD chain (MZM, optical fiber, photodiode) is replaced by a physics-inspired neural-network surrogate whose structure mirrors the device physics, rather than a generic feedforward layer [16, 17]. With the surrogates frozen, a neural Tx-side pre-distorter and a neural Rx-side feed-forward equalizer are jointly trained by back-propagating the loss through this differentiable chain in a direct-learning architecture [18]. The second is a transformer-based neural-pulse-shaping transceiver in which the fixed Nyquist pulse is replaced by an attention-based module [19]: NN_{Tx} shapes the transmitted pulse at four-times oversampling, while a matched receiver module NN_{Rx} performs symbol-level detection through an argmax over BPAM amplitude classes. The pair is trained jointly with a cross-entropy loss across a wide signal-to-noise-ratio (SNR) range, so that the effective received pulse (the cascade of NN_{Tx} , the band-limited transmitter, the square-law detector and NN_{Rx}) is essentially free of inter-symbol interference and approaches the optimal transceiver for the BPAM/DD additive-white-Gaussian-noise (AWGN) channel. Related learned-pulse-shaping strategies have recently been explored in other system contexts [20].

Demonstration Architecture and Experimental Setup

The demonstration compares the optimization strategies of Section 2 on a single short-reach IM/DD link (Fig. 1). A control-and-visualization laptop drives both a local simulation engine and, over a live connection, a remotely operated optical experiment in our laboratory; a monitor at the booth displays waveforms, eye diagrams, and bit-error-rate (BER) versus SNR metrics, with posters describing the underlying methods.

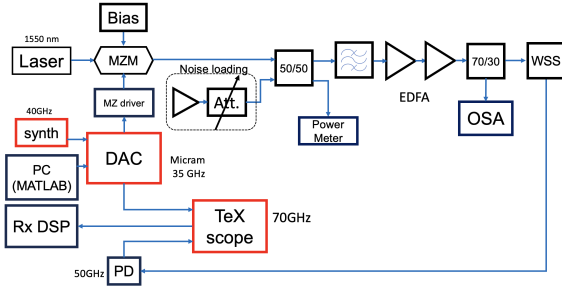


Fig. 2: Experimental setup of the demonstration.

Training and deployment are deliberately decoupled. All models are trained offline (surrogate-based for the pre-distorter and the end-to-end transceivers, supervised on captured data for the neural-network equalizer, and direct table population for the LUT) while the live link executes only the frozen, deployed models, with an animated side panel walking attendees through the training workflow.

In the local demonstration, the interface runs the methods on a calibrated digital surrogate of the complete IM/DD link. The workflow proceeds from waveform generation through synchronization, re-sampling, receiver DSP, and BER estimation under controlled noise loading. The transmitter-side pre-distortion, the three receiver-side methods (adaptive equalization, neural-network equalization, and LUT detection), and the two end-to-end models can be enabled individually or in combination.

In the remote-laboratory component, the identical workflow runs on physical hardware (Fig. 2) [5]. A 1550-nm continuous-wave external-cavity laser is intensity-modulated by an MZM held at its null bias for BPAM operation and driven, through an electrical driver, by a MICRAM DAC at 80 GSa/s with 35-GHz analog bandwidth, using twofold-oversampled raised-cosine pulse shaping with roll-off 0.85. Amplified spontaneous emission (ASE) noise is loaded via an erbium-doped fiber amplifier (EDFA), a variable optical attenuator, and an analog optical bandpass filter, setting the optical signal-to-noise ratio (OSNR); the signal is then preamplified by a second EDFA and filtered by a programmable optical waveshaper at the input of a 50-GHz photodiode, whose output is digitized by a 70-GHz real-time oscilloscope at 200 GSa/s before receiver DSP. A waveform generated by the local software is uploaded to the transmitter, transmitted, acquired, and processed through the selected DSP pipeline, so the audience sees not only the final performance curves but the full operational sequence from waveform generation to data-driven DSP validation.

In the live configuration, attendees choose a method, trigger a remote transmission, and watch the resulting waveforms, eye diagrams, and BER/SNR update in real time.

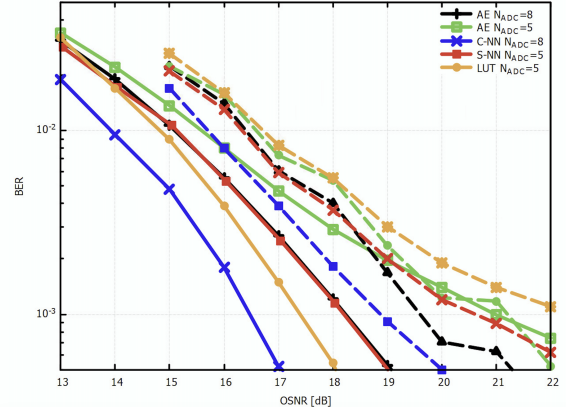


Fig. 3: BER versus SNR.

Validation and Demonstration Outputs

Each technique is evaluated and displayed with a common set of metrics. The system is operated at a fixed symbol rate while injected ASE noise sweeps the per-acquisition SNR, and BER is measured for every strategy on identical captured sequences; eye diagrams accompany each operating point as a visual confirmation.

In simulation (Fig. 3), the deeper neural-network equalizer provides the largest gain over the adaptive equalizer, about 1 dB at $\text{BER} = 10^{-2}$ and approaching 2 dB at 10^{-3} , with the LUT detector reaching the same behavior at negligible inference cost; experiments reproduce the simulation trends with a residual implementation penalty of the order of 2 dB, with the LUT additionally bounded by the size of the experimental training set [5]. The fairly trained memory polynomial pre-distorter recovers part of this penalty by linearizing the transmitter front-end. The transformer neural-pulse-shaping transceiver, evaluated on the same platform, delivers up to 5.3 dB of SNR improvement over a root-raised-cosine pulse of equivalent bandwidth.

Conclusions

A single 20-GBd BPAM IM/DD platform is used to compare the three families of AI-assisted optimization for short-reach optics under identical experimental conditions: a fairly trained memory polynomial pre-distorter at the transmitter, neural-network and LUT equalization at the receiver, and a transformer-based neural-pulse-shaping transceiver acting end-to-end. This unified experimental implementation on BPAM, a format on which such a side-by-side comparison has not previously been reported, is the novelty achieved by the proposed demonstration.

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